Titanic Dataset Analysis

Business Scenario

The year is **1912**. Following the sinking of the RMS Titanic, the **White Star Line** has commissioned your data analytics team to investigate the passenger records.

The company wants to understand:

- What characteristics are associated with passenger survival?
- Are there any data quality issues that must be addressed before predictive modeling?

Question 1: Submission Requirements

Part 1: Analysis Report

- File name: StudentID_hw_2_1.pdf
- Content: Complete answers to all questions, including explanations, interpretations, and visualizations.

Part 2: Source Files

- Submit all source files used in the analysis: File name: StudentID_hw_2_1.ipynb (or .xlsx)

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Question 2: Data Quality Assessment

Q2.1: Missing Data Analysis

1. Check for missing values in the dataset:
    - Which columns contain missing values?
    - What percentage of missing values does each column have?
    - Is it appropriate to simply delete the Cabin column? Explain your reasoning.

2. Create a missing data heatmap using an appropriate visualization:
    - What patterns can you observe from the heatmap?
    - Are there relationships between missing values across different columns?

Q2.2: Outlier Detection

1. Identify outliers in 2–3 numerical variables using appropriate methods (e.g.,

IQR method, box plots, or z-score).
2. Why was the IQR method chosen instead of the Z-score method?
3. Are all detected outliers actually "bad data"? Give examples from the Titanic dataset
4. Should Fare outliers be removed before building a predictive model? Justify your answer.
5. Besides the IQR method, suggest one additional outlier detection technique and discuss its advantages.
6. Discuss the potential impact of these outliers on your analysis.

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Question 3: Univariate Analysis with Visualization

Q3.1: Numerical Variables

1. Create histograms for the columns Age and Fare:
    - What do the histograms reveal about the distribution of these variables?
    - Are the distributions normal, skewed, or do they have other characteristics?
2. Create box plots for Age, Fare, and SibSp (number of siblings/spouses):
    - What do the box plots indicate about distribution and outliers?
    - Compare the distributions and central tendencies of these variables.

Q3.2: Categorical Variables

1. Analyze the distribution of each categorical variable (e.g., Sex, Pclass, Embarked, Survived).
2. Create appropriate visualizations (bar charts, pie charts).
3. Summarize the key findings related to the distribution of categorical variables.

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Question 4: Bivariate Analysis – Numerical Variables

Q4.1: Correlation Analysis

1. Compute the Pearson correlation coefficient between Age and Fare:
    - Interpret the correlation value.
    - Is the relationship strong, moderate, or weak?
2. Create a scatter plot showing the relationship between Age and Fare:
    - What patterns do you observe?
    - Are there clusters or interesting trends?

Question 5: Bivariate Analysis – Categorical Variables

Q5.1: Contingency Table Analysis

Create a contingency table showing the relationship between Survival and Pclass:

- What does the table reveal about survival rates across different classes?
- Compute the survival percentage for each class.
- What conclusion can you draw about the relationship between these variables?

Q5.2: Statistical Testing

Perform a Chi-Square Test of Independence between Survival and Pclass:

- State the null hypothesis and alternative hypothesis.
- Report the test statistic, p-value, and degrees of freedom.
- What conclusion can you draw about the relationship between these variables?

Question 6: Multi-variate Analysis

Q6.1: Create a pairplot for the variables Age, Fare, and SibSp:

- What deeper insights can you gain from this pairplot?
- Identify any interesting relationships between the variables.
- Report any outliers from this pairplot